MONEYRULES SERIES

Compound Interest

The Eighth Wonder of the World

The formula, the psychology, and worked scenarios for retirement, children's savings and house deposits.

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1. Introduction — Why Einstein Called It a Wonder

"Compound interest is the eighth wonder of the world. He who understands it, earns it. He who doesn't, pays it." The quote is usually attributed to Einstein, though historians are unsure. What is certain is that compound interest is the most important single concept in personal finance — yet most people profoundly underestimate its power.

The reason it feels magical is that it is EXPONENTIAL, and humans are evolved to think linearly. A savings account paying 7% per year doesn't grow 7% every year from the original balance — it grows 7% of the ever-larger total, including past interest. Each year's interest itself starts earning interest.

> Simple interest grows in a line.
> Compound interest grows in a curve.
> Over decades, the curve wins by miles.

2. The Formula, Broken Down

The compound interest formula is:

$$A = P \times (1 + r/n)^{(n \cdot t)}$$

Where:

- A = Final amount
- P = Principal (starting amount)
- r = Annual interest rate (as a decimal, e.g. 0.05 for 5%)
- n = Compoundings per year (12 = monthly, 365 = daily)
- t = Time in years

A Manual Calculation

You deposit £10,000 at 6% annually, compounded monthly, for 20 years. $A = 10{,}000 \times (1 + 0.06/12)^{(12 \times 20)} = 10{,}000 \times (1.005)^{240} \approx$ £33,102.

You put in £10,000. You end with £33,102. Over £23,000 of pure growth, from doing nothing but waiting.

3. Simple vs Compound: The Critical Difference

Consider £10,000 invested at 7% for 40 years:

Method	Final Value	Growth
Simple interest	£38,000	3.8× starting amount
Compound interest (annual)	£149,744	14.9× starting amount

Same money. Same rate. Same time period. Four times the final wealth — purely because compound re-invests the interest to earn more interest.

> The difference between simple and compound
> is the difference between middle-class and wealthy.

4. The Power of TIME (Not Amount)

This may be the single most important financial chart of your life. Two investors, both retire at 65. They earn the same 7% annual return.

Investor	Saves from age...	Total invested	Value at 65
Anna	25 to 35 (10 yrs), then stops	£24,000 (£200/mo × 10 yrs)	£266,000
Ben	35 to 65 (30 yrs)	£72,000 (£200/mo × 30 yrs)	£245,000

Anna invested only £24,000 and stopped. Ben invested THREE TIMES as much, for THREE TIMES as long. Anna still ends up with more.

Why? Anna's money had 10 extra years of compounding. Those 10 years are worth more than the next 30 combined.

> Starting early beats investing more.
> Every single time.

5. The Effect of RATE: Why 1% Matters

£10,000 invested for 30 years at different rates:

Rate	Final Value	Extra vs 4%
3%	£24,273	—
4%	£32,434	Baseline
5%	£43,219	+£10,785
6%	£57,435	+£25,001
7%	£76,123	+£43,689
8%	£100,627	+£68,193

A 1% higher rate over 30 years adds £10,000+ on just £10k starting capital. This is why paying a 1.5% fund management fee instead of 0.2% is catastrophic over a career.

> Low-cost index funds keep your 1% in your pocket.
> Over 30 years, that 1% becomes hundreds of thousands.

6. The Effect of FREQUENCY (Annual/Monthly/Daily)

£10,000 at 6% for 20 years, compounded at different intervals:

Compounding	Formula Effect	Final Value
Annually (n=1)	(1.06)^20	£32,071
Quarterly (n=4)	(1.015)^80	£32,907
Monthly (n=12)	(1.005)^240	£33,102
Daily (n=365)	(1.000164)^7,300	£33,198
Continuous	e^(0.06 × 20)	£33,201

More-frequent compounding helps, but with diminishing returns. Moving from annual to monthly is worth about £1,000 here. Moving from monthly to daily is worth only £100 more. Don't obsess over this — rate and time matter far more.

7. The Reverse Calculator: Saving for a Goal

Often the useful question is not "how much will I have?" but "how much should I save?"

Formula for Monthly Saving Needed

$$PMT = FV \times r / ((1+r)^n - 1)$$

Where FV = target value, r = monthly rate, n = number of months.

Worked Example

You want £100,000 in 20 years. Assumed return: 7% annual (0.583% monthly).

$PMT = 100{,}000 \times 0.00583 / ((1.00583)^{240} - 1) \approx £192/month$

> £192/month for 20 years → £100,000.
> You've contributed £46,000 of it. Compounding added £54,000.

8. Real Scenarios: Retirement, Kids, House Deposit

Retirement at 65 starting at 30

£300/month into a pension at 7% from age 30 to 65 = £543,000. The same £300/month from age 45 to 65 = only £156,000. 15 extra years = 3.5× more wealth.

University fund for a newborn

£100/month in a Junior ISA at 6% for 18 years = £38,929. £21,600 of that is your contribution — £17,329 is pure compounding growth.

House deposit in 7 years

Target £30,000 in 7 years at 5% = £300/month. At 3% (cash savings only) you'd need £325/month. A Lifetime ISA with its 25% government bonus turns £300/month into your full £30,000 in ~5 years.

> Pair compounding with government-boosted accounts
> (Pension relief, LISA bonus)
> — free money on top of already-exponential growth.

9. The Dark Side: When Compound Interest Works Against You

The same mathematical force that grows your wealth grows your debt — only faster, because consumer debt rates are typically far higher than investment returns.

Debt Type	Typical APR	£1,000 after 10 yrs unpaid
Mortgage	4–6%	£1,480 – £1,790
Personal loan	8–15%	£2,160 – £4,046
Credit card	20–30%	£6,192 – £13,786
Payday loan	400%+	Catastrophic

Minimum-payment credit card debt is financial quicksand. The bank is compounding YOUR money into THEIR pocket, often at rates 3–5× faster than your investments can grow. Kill high-interest debt before investing.

> The question isn't whether compound interest will shape your life.
> It's whether it will shape it for you or against you.

10. Key Takeaways and Action Plan

- Compound = interest earning interest. It's exponential, not linear.
- TIME matters more than AMOUNT — start as early as possible.
- 1% more return can mean hundreds of thousands of pounds over a career.
- Monthly/daily compounding is nice but not critical. Rate + time are what matter.
- Use the reverse formula to work out how much to save for a goal.
- Clear high-interest debt FIRST — it compounds against you faster than investments compound for you.
- Pair compounding with tax wrappers (ISA, SIPP, LISA) for double benefit.

Your First Week

1. Day 1: Calculate what £100/month at 7% for YOUR remaining years to 65 would produce.
2. Day 2: Open a Stocks & Shares ISA if you don't have one (~10 minutes).
3. Day 3: Set up a monthly direct debit — even £50 is a start.
4. Day 4: Review any credit card debt above 15% APR — attack this first.
5. Day 5: Pick an ACCUMULATION share class (re-invests dividends automatically).
6. Day 6: Diarise a quarterly review to increase contributions as income rises.
7. Day 7: Leave it alone. Seriously. The magic happens in the decades, not days.

Visit www.incomeonline.info for 199+ ways to grow the "invest more each month" side of the equation.

Thank you for reading.

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